

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

FORMAL DIGITAL FORENSICS INVESTIGATION REPORT

SBT-DF203 - LAB 7: DNS Introduction and Traffic Analysis

Case Identifier	SBT-DF203-Lab7
Lead Examiner	Nebeuwa Ifeanyichukwu Raphael
Registration / ID No.	202520850LE
Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 7
Lab Title	DNS Introduction and Traffic Analysis
Date of Submission	September 22, 2026
Primary Target System	Kali Linux Workstation (kali@kali)
Evidence Type	DNS PCAP/PCAPNG captures and associated analysis artifacts

Prepared for authorized academic forensic examination

1. Executive Summary

This report documents the completion of SBT-DF203 Lab 7, a DNS-focused network-forensics exercise conducted in a Kali Linux analysis environment. The practical established the lab workspace, identified the configured DNS resolver, preserved and hashed the supplied DNS capture, queried A/AAAA/MX/NS records with dig, captured fresh DNS traffic with TShark, extracted DNS query and response fields, matched transaction IDs, and examined DNS generated by a browser visit to example.com.

The observed resolver for the captured traffic was 192.168.146.2 on UDP port 53, with the Kali client using 192.168.146.132. The A query for example.com returned two IPv4 addresses, 172.66.147.243 and 104.20.23.154, each with a TTL of 5 seconds in the captured response. The AAAA query returned two IPv6 addresses, also with TTL values of 5 seconds. The fresh DNS correlation showed transaction IDs 0xf0ba for the A query and 0x4e43 for the AAAA query.

Part F was completed to the extent supported by the evidence. The DNS A-answer extraction identified 172.66.147.243 and 104.20.23.154, but the subsequent TCP-destination extraction returned no rows. This is consistent with the browser capture having been restricted to port 53 DNS traffic, so later TCP/HTTPS packets were not present in that PCAPNG. Part G, the SMTP DNS analysis, was not performed because the lab instructions identify it as optional.

The supplied dig_dns.pcap capture and working copy matched at SHA-256 9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d. The locally generated fresh_dig_dns.pcapng capture was hashed as c0a8de397bd413a19f35ec601cc2f8d84bc879ef16b1e401209a891439ac46d9.

2. Investigation Overview and Scope

2.1 Purpose and Objectives

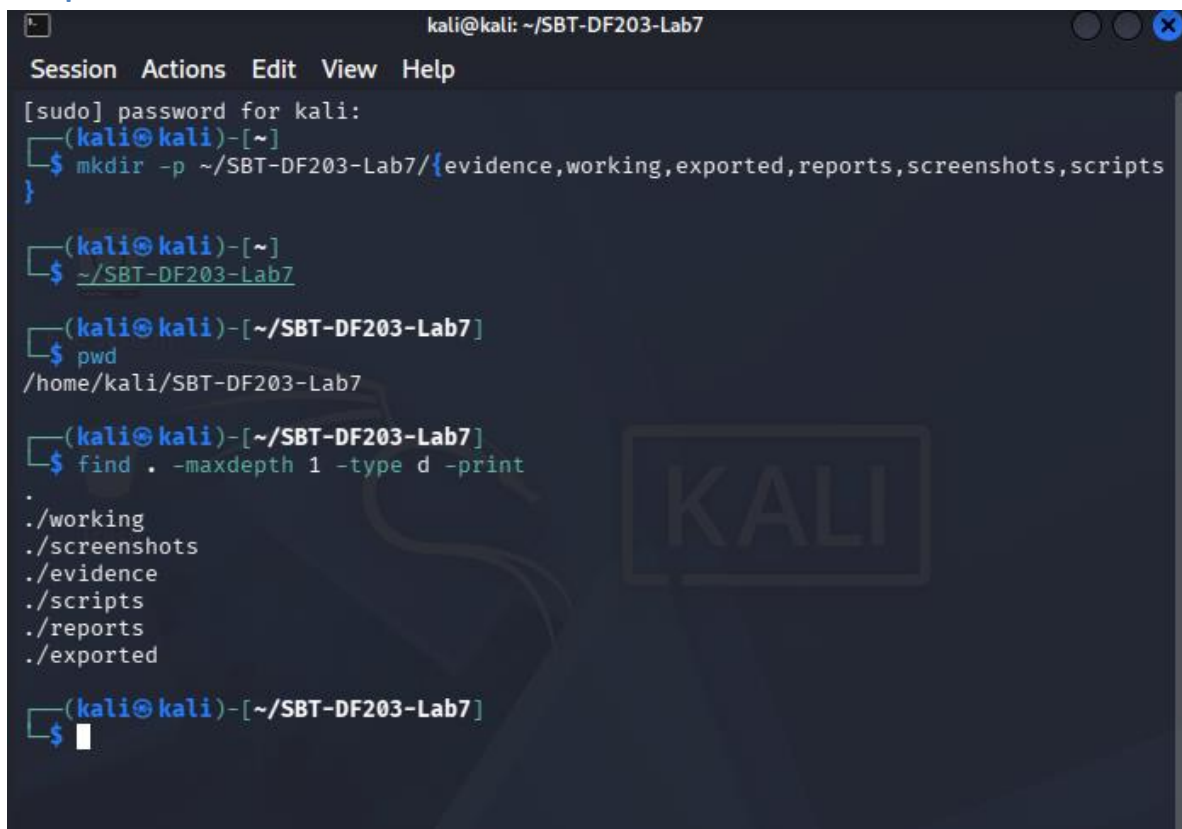
- Identify the configured DNS resolver and document the resolver configuration.
- Preserve and hash the supplied DNS evidence before analysis.
- Use dig to query A, AAAA, MX and NS records for example.com.
- Capture a controlled fresh DNS query and preserve the resulting PCAPNG.
- Extract DNS transaction IDs, query names, record types, response codes, answers and TTL values.
- Match DNS queries to responses using transaction ID and endpoint information.
- Analyze DNS generated by a browser visit and inventory observed query names.
- Correlate DNS answers with later connections where the capture contains the necessary transport-layer evidence.
- Document limitations and normal DNS variations without treating them as anomalies.

2.2 Environment and Tools

Component	Observed configuration
Operating System	Kali Linux
Analysis directory	~/SBT-DF203-Lab7

Client IP observed	192.168.146.132
DNS resolver	192.168.146.2
DNS transport	UDP/53
Interface used for live capture	eth0
DNS utility	dig
Capture/analysis	TShark / Wireshark
Primary domain tested	example.com

2.3 Workspace and Resolver Evidence



```
kali@kali: ~/SBT-DF203-Lab7
Session Actions Edit View Help
[sudo] password for kali:
(kali@kali)-[~]
$ mkdir -p ~/SBT-DF203-Lab7/{evidence,working,exported,reports,screenshots,scripts}
(kali@kali)-[~]
$ ~/SBT-DF203-Lab7
(kali@kali)-[~/SBT-DF203-Lab7]
$ pwd
/home/kali/SBT-DF203-Lab7
(kali@kali)-[~/SBT-DF203-Lab7]
$ find . -maxdepth 1 -type d -print
.
./working
./screenshots
./evidence
./scripts
./reports
./exported
(kali@kali)-[~/SBT-DF203-Lab7]
$
```

Figure 1: Lab 7 workspace directories created under ~/SBT-DF203-Lab7.

```
(kali㉿kali)-[~/SBT-DF203-Lab7]
$ cat /etc/resolv.conf | tee reports/resolv_conf.txt
# Generated by NetworkManager
search localdomain
nameserver 192.168.146.2

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ resolvectl status 2>/dev/null | tee reports/resolvectl_status.txt || true

(kali㉿kali)-[~/SBT-DF203-Lab7]
$
```

Figure 2: Resolver configuration recorded from `/etc/resolv.conf`; nameserver 192.168.146.2 was observed.

```
kali@kali: ~/SBT-DF203-Lab7
Session Actions Edit View Help

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ wget -O evidence/dig_dns.pcap \
'https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Fo
rensics/lab_files/dns/dig_dns.pcap'
--2026-09-21 01:30:34-- https://raw.githubusercontent.com/frankwxu/digital-forensic
s-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com) ... 185.199.108.133,
185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|
:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1804656 (1.7M) [application/octet-stream]
Saving to: 'evidence/dig_dns.pcap'

evidence/dig_dns.pca 100%[=====>] 1.72M 1.69MB/s in 1.0s
2026-09-21 01:30:36 (1.69 MB/s) - 'evidence/dig_dns.pcap' saved [1804656/1804656]

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ cp --preserve=timestamps evidence/dig_dns.pcap working/dig_dns_working.pcap

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap | tee reports/dns_c
apture_hashes.txt
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d evidence/dig_dns.p
cap
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d working/dig_dns_wo
rking.pcap

(kali㉿kali)-[~/SBT-DF203-Lab7]
$
```

Figure 3: Supplied `dig_dns.pcap` downloaded, copied to the working directory, and hashed.

3. Evidence Preservation and Integrity

The supplied capture `evidence/dig_dns.pcap` was preserved and copied to `working/dig_dns_working.pcap` before analysis. The recorded SHA-256 digest for both files was identical, supporting bit-for-bit consistency between the preserved source and working copy.

Evidence / Artifact	Role	SHA-256
evidence/dig_dns.pcap	Supplied/original DNS evidence	9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d
working/dig_dns_working.pcap	Analysis working copy	9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d
evidence/fresh_dig_dns.pcapng	Locally generated fresh DNS evidence	c0a8de397bd413a19f35ec601cc2f8d84bc879ef16b1e401209a891439ac46d9

The fresh capture was initially attempted with a root-owned TShark process writing directly to the evidence directory. That attempt produced a permission error. A corrected workflow wrote the capture to /tmp, after which the file was moved into evidence/ and its ownership changed to the Kali user for analysis. The corrected capture was successfully hashed and retained.

```
(kali㉿kali)-[~/SBT-DF203-Lab7]
└─$ IFACE=eth0
sudo tshark -i "$IFACE" -f 'port 53' -a duration:40 -w /tmp/browser_dns.pcapng &
[3] 628157

(kali㉿kali)-[~/SBT-DF203-Lab7]
└─$ Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
6

[3] done sudo tshark -i "$IFACE" -f 'port 53' -a duration:40 -w /tmp/browser_dns.pcapn
(kali㉿kali)-[~/SBT-DF203-Lab7]
└─$ ls -lh /tmp/browser_dns.pcapng
-rw-r--r-- 1 root root 1.1K Sep 22 11:40 /tmp/browser_dns.pcapng

(kali㉿kali)-[~/SBT-DF203-Lab7]
└─$ sudo mv /tmp/browser_dns.pcapng evidence/browser_dns.pcapng

(kali㉿kali)-[~/SBT-DF203-Lab7]
└─$ sudo chown kali:kali evidence/browser_dns.pcapng

(kali㉿kali)-[~/SBT-DF203-Lab7]
└─$
```

Figure 4: Corrected fresh DNS capture workflow completed on eth0 and the PCAPNG was moved into evidence/.

```

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ sudo mv /tmp/fresh_dig_dns.pcapng evidence/fresh_dig_dns.pcapng

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ sudo chown kali:kali evidence/fresh_dig_dns.pcapng

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ ls -lh evidence/fresh_dig_dns.pcapng
-rw----- 1 kali kali 844 Sep 21 01:55 evidence/fresh_dig_dns.pcapng

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ sha256sum evidence/fresh_dig_dns.pcapng | tee reports/fresh_dns_sha256.txt
c0aa8de397bd413a19f35ec601cc2f8d84bc879ef16b1e401209a891439ac46d evidence/fresh_dig_dns.pcapng

(kali㉿kali)-[~/SBT-DF203-Lab7]
$

```

Figure 5: SHA-256 hash recorded for evidence/fresh_dig_dns.pcapng.

4. Part A - DNS Record Queries with dig

The required A, AAAA, MX and NS queries were executed against example.com. The resolver shown by the dig output was 192.168.146.2#53 over UDP. The responses generally returned NOERROR. The MX and NS commands each displayed an initial timeout message before producing a successful DNS response.

```

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ dig example.com A | tee reports/dig_example_A.txt

; <<>> DiG 9.20.27-2-Debian <<>> example.com A
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 64369
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                 5       IN      A      172.66.147.243
example.com.                 5       IN      A      104.20.23.154

;; Query time: 331 msec
;; SERVER: 192.168.146.2#53(192.168.146.2) (UDP)
;; WHEN: Mon Sep 21 01:33:47 EDT 2026
;; MSG SIZE rcvd: 72

```

Figure 6: dig example.com A output showing two IPv4 answers and resolver 192.168.146.2.

```

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ dig example.com AAAA | tee reports/dig_example_AAAA.txt

; <<>> DiG 9.20.27-2-Debian <<>> example.com AAAA
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 7423
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 4096
;; QUESTION SECTION:
;example.com.                IN      AAAA

;; ANSWER SECTION:
example.com.                5       IN      AAAA    2606:4700:10::ac42:93f3
example.com.                5       IN      AAAA    2606:4700:10::6814:179a

;; Query time: 279 msec
;; SERVER: 192.168.146.2#53(192.168.146.2) (UDP)
;; WHEN: Mon Sep 21 01:33:55 EDT 2026
;; MSG SIZE rcvd: 96

```

Figure 7: dig example.com AAAA output showing two IPv6 answers.

```

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ dig example.com MX | tee reports/dig_example_MX.txt
;; communications error to 192.168.146.2#53: timed out

; <<>> DiG 9.20.27-2-Debian <<>> example.com MX
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 36065
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      MX

;; ANSWER SECTION:
example.com.                5       IN      MX      0 .

;; Query time: 1123 msec
;; SERVER: 192.168.146.2#53(192.168.146.2) (UDP)
;; WHEN: Mon Sep 21 01:34:11 EDT 2026
;; MSG SIZE rcvd: 55

```

Figure 8: dig example.com MX output. The displayed successful response contains MX preference 0 and exchange '.'.

```

Session  Actions  Edit  View  Help

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ dig example.com NS | tee reports/dig_example_NS.txt
;; communications error to 192.168.146.2#53: timed out

; <<>> DiG 9.20.27-2-Debian <<>> example.com NS
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 58852
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      NS

;; ANSWER SECTION:
example.com.                 5       IN      NS      elliot.ns.cloudflare.com.
example.com.                 5       IN      NS      hera.ns.cloudflare.com.

;; Query time: 1128 msec
;; SERVER: 192.168.146.2#53(192.168.146.2) (UDP)
;; WHEN: Mon Sep 21 01:34:38 EDT 2026
;; MSG SIZE rcvd: 95

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ dig +short example.com A | tee reports/dig_example_short.txt
172.66.147.243
104.20.23.154

(kali㉿kali)-[~/SBT-DF203-Lab7]
$

```

Figure 9: dig example.com NS output and the subsequent dig +short A result.

Query	Status	Answers	Transaction ID	Query time	TTL	Observed answer
A	NOERROR	2	64369	331 ms	5	172.66.147.243; 104.20.23.154
AAAA	NOERROR	2	7423	279 ms	5	2606:4700:10::ac42:93f3; 2606:4700:10::6814:179a
MX	NOERROR	1	36065	1123 ms	5	0 .
NS	NOERROR	2	58852	1128 ms	5	elliott.ns.cloudflare.com; hera.ns.cloudflare.com

4.1 Record-Type Interpretation

- IN is the Internet DNS class.
- A maps a hostname to an IPv4 address.
- AAAA maps a hostname to an IPv6 address.
- MX identifies mail-exchange information for a domain, including a preference value and mail-exchange target.
- NS identifies the authoritative name servers associated with a DNS zone.
- TTL is the amount of time a DNS answer may remain cached under normal DNS caching rules; it is not the creation date of the record.

5. Part B - Fresh DNS Capture

A fresh DNS capture was generated on eth0 using a 25-second TShark capture filtered to port 53. After the capture was running, controlled dig queries for example.com were generated. The successful capture was preserved as evidence/fresh_dig_dns.pcapng and hashed.

```

Session  Actions  Edit  View  Help

(kali@kali)-[~/SBT-DF203-Lab7]
$ sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w /tmp/fresh_dig_dns.pcapng &
[2] 31839

(kali@kali)-[~/SBT-DF203-Lab7]
$ Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
dig +noedns example.com A >/dev/null
1

(kali@kali)-[~/SBT-DF203-Lab7]
2 dig +noedns example.com AAAA >/dev/null

(kali@kali)-[~/SBT-DF203-Lab7]
4 $

[2] done sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w
(kali@kali)-[~/SBT-DF203-Lab7]
$ dig +noedns example.com MX >/dev/null

(kali@kali)-[~/SBT-DF203-Lab7]
$ wait

(kali@kali)-[~/SBT-DF203-Lab7]
$ ls -lh /tmp/fresh_dig_dns.pcapng
-rw-r--r-- 1 root root 844 Sep 21 01:55 /tmp/fresh_dig_dns.pcapng

(kali@kali)-[~/SBT-DF203-Lab7]
$

```

Figure 10: Successful fresh DNS capture and subsequent controlled dig traffic generation.

```

(kali@kali)-[~/SBT-DF203-Lab7]
$ sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w /tmp/fresh_dig_dns.pcapng &
[2] 29519

(kali@kali)-[~/SBT-DF203-Lab7]
$ Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'

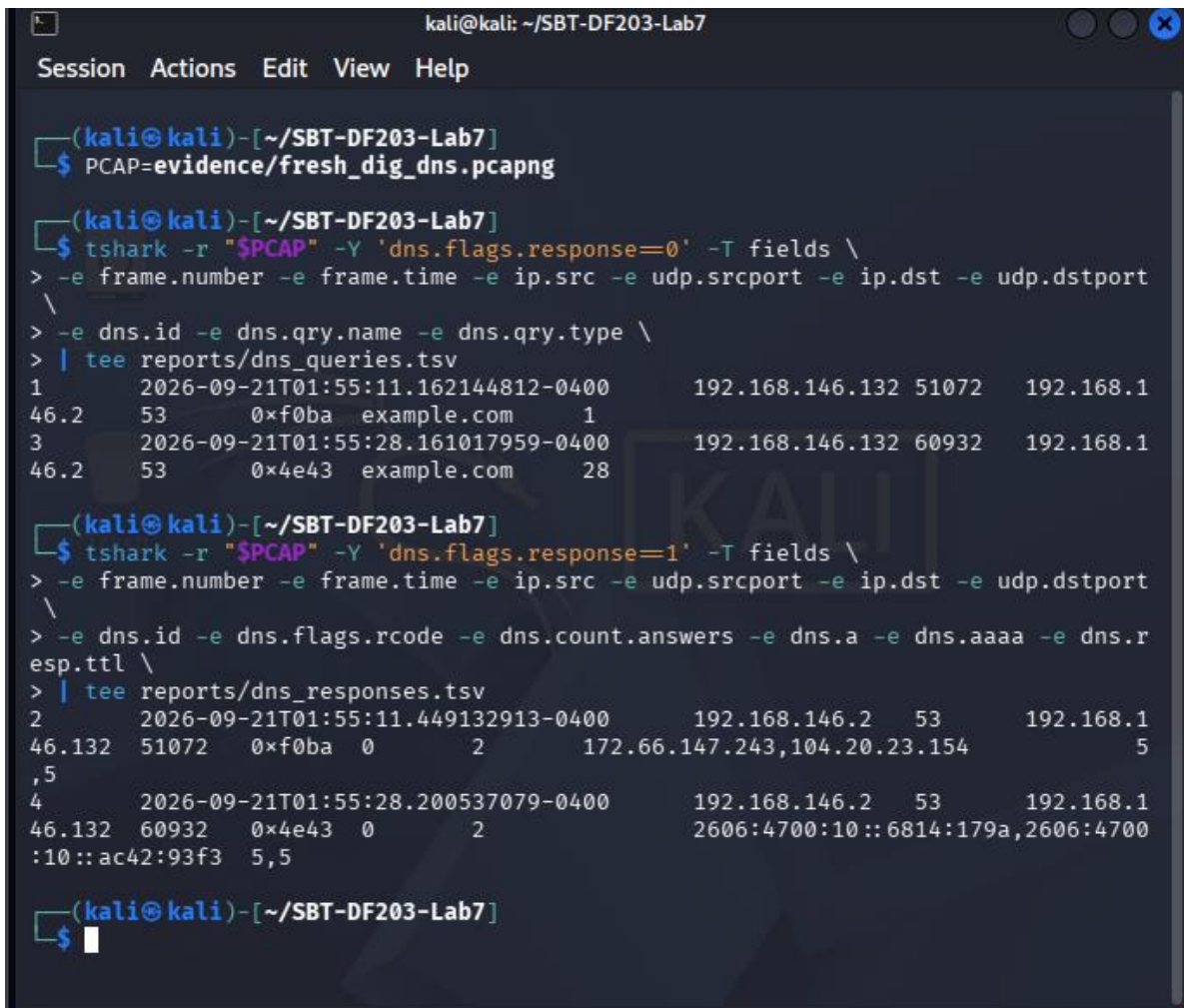
[2] done sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w

```

Figure 11: TShark completed the fresh capture on eth0.

An earlier attempt to save directly into the evidence directory failed with a permission-denied error. The final capture was written to /tmp and then moved into evidence/, after which ownership and permissions were corrected. This operational correction is recorded because evidence handling and provenance are part of the forensic workflow.

5.1 Part C - Extracted DNS Query and Response Fields



```

kali@kali: ~/SBT-DF203-Lab7
Session Actions Edit View Help

(kali@kali)-[~/SBT-DF203-Lab7]
$ PCAP=evidence/fresh_dig_dns.pcapng

(kali@kali)-[~/SBT-DF203-Lab7]
$ tshark -r "$PCAP" -Y 'dns.flags.response==0' -T fields \
> -e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
> -e dns.id -e dns.qry.name -e dns.qry.type \
> | tee reports/dns_queries.tsv
1      2026-09-21T01:55:11.162144812-0400      192.168.146.132 51072      192.168.1
46.2    53      0xf0ba example.com      1
3      2026-09-21T01:55:28.161017959-0400      192.168.146.132 60932      192.168.1
46.2    53      0x4e43 example.com      28

(kali@kali)-[~/SBT-DF203-Lab7]
$ tshark -r "$PCAP" -Y 'dns.flags.response==1' -T fields \
> -e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
> -e dns.id -e dns.flags.rcode -e dns.count.answers -e dns.a -e dns.aaaa -e dns.r
esp.ttl \
> | tee reports/dns_responses.tsv
2      2026-09-21T01:55:11.449132913-0400      192.168.146.2 53      192.168.1
46.132 51072 0xf0ba 0      2      172.66.147.243,104.20.23.154      5
,5
4      2026-09-21T01:55:28.200537079-0400      192.168.146.2 53      192.168.1
46.132 60932 0x4e43 0      2      2606:4700:10::6814:179a,2606:4700
:10::ac42:93f3 5,5

(kali@kali)-[~/SBT-DF203-Lab7]
$

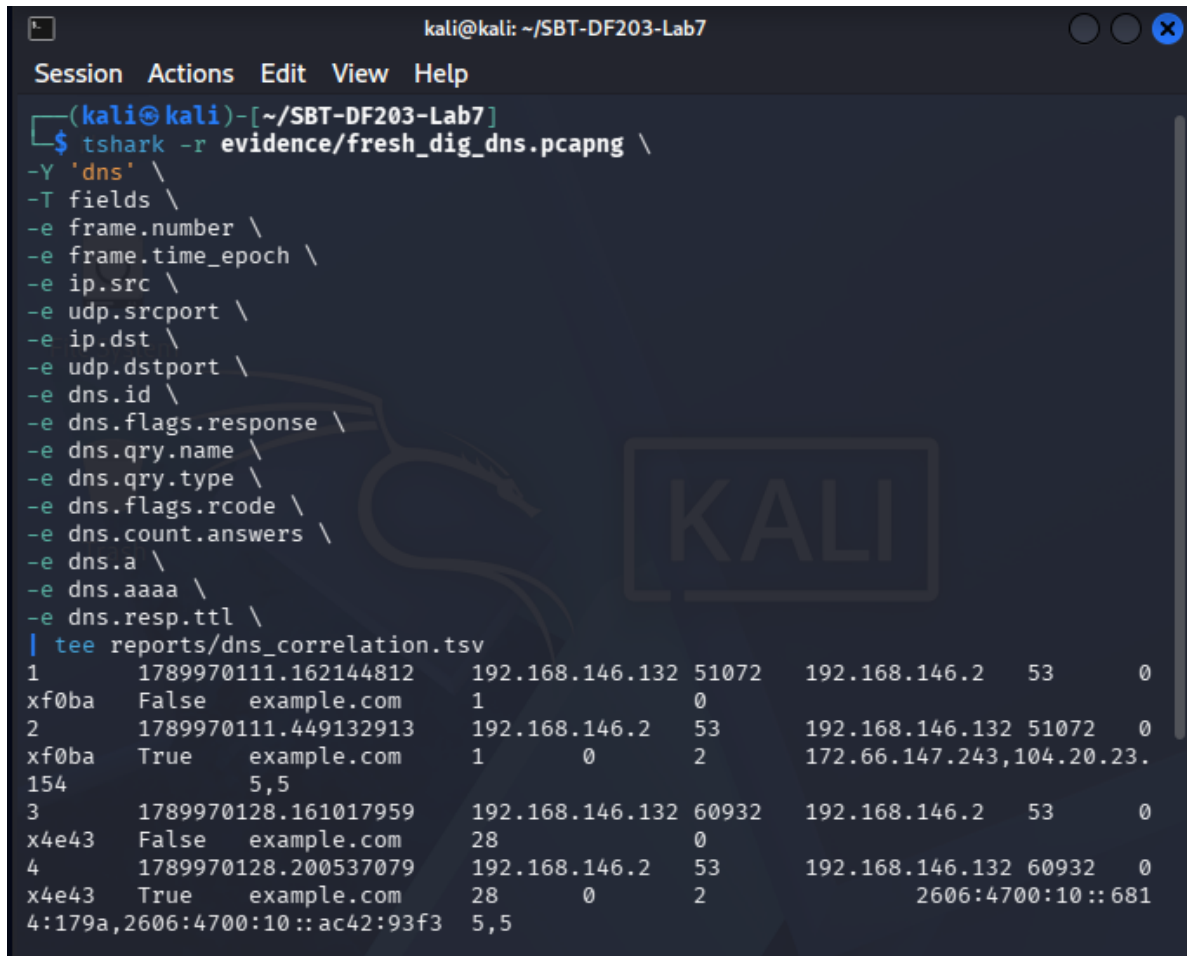
```

Figure 12: TShark extraction of DNS query and response fields from fresh_dig_dns.pcapng.

The fresh capture contained an A query and an AAAA query. The A query originated from 192.168.146.132:51072 and was sent to resolver 192.168.146.2:53. Its transaction ID was 0xf0ba. The response returned code 0 (NOERROR), two A answers and TTL values of 5 seconds.

The AAAA query originated from 192.168.146.132:60932 and used transaction ID 0x4e43. The response again returned code 0, two AAAA answers and TTL values of 5 seconds.

6. Part D - Transaction Matching and Timing



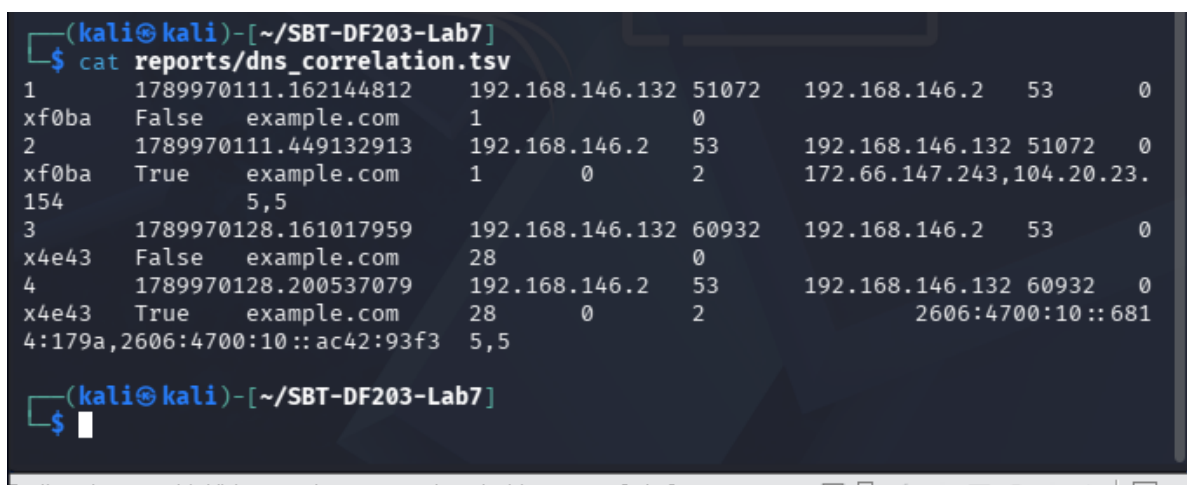
```

kali@kali: ~/SBT-DF203-Lab7
Session Actions Edit View Help

(kali@kali)-[~/SBT-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng \
-Y 'dns' \
-T fields \
-e frame.number \
-e frame.time_epoch \
-e ip.src \
-e udp.srcport \
-e ip.dst \
-e udp.dstport \
-e dns.id \
-e dns.flags.response \
-e dns.qry.name \
-e dns.qry.type \
-e dns.flags.rcode \
-e dns.count.answers \
-e dns.a \
-e dns.aaaa \
-e dns.resp.ttl \
| tee reports/dns_correlation.tsv
1      1789970111.162144812      192.168.146.132 51072      192.168.146.2 53      0
xf0ba  False  example.com 1      0
2      1789970111.449132913      192.168.146.2 53      192.168.146.132 51072 0
xf0ba  True   example.com 1      0      2      172.66.147.243,104.20.23.
154    5,5
3      1789970128.161017959      192.168.146.132 60932      192.168.146.2 53      0
x4e43  False  example.com 28     0
4      1789970128.200537079      192.168.146.2 53      192.168.146.132 60932 0
x4e43  True   example.com 28     0      2      2606:4700:10::681
4:179a,2606:4700:10::ac42:93f3 5,5

```

Figure 13: Transaction-level DNS correlation showing matching query/response transaction IDs, endpoints, answers and TTLs.



```

(kali@kali)-[~/SBT-DF203-Lab7]
$ cat reports/dns_correlation.tsv
1      1789970111.162144812      192.168.146.132 51072      192.168.146.2 53      0
xf0ba  False  example.com 1      0
2      1789970111.449132913      192.168.146.2 53      192.168.146.132 51072 0
xf0ba  True   example.com 1      0      2      172.66.147.243,104.20.23.
154    5,5
3      1789970128.161017959      192.168.146.132 60932      192.168.146.2 53      0
x4e43  False  example.com 28     0
4      1789970128.200537079      192.168.146.2 53      192.168.146.132 60932 0
x4e43  True   example.com 28     0      2      2606:4700:10::681
4:179a,2606:4700:10::ac42:93f3 5,5

```

Figure 14: Saved dns_correlation.tsv output used for the query-response matching analysis.

TX ID	Query	Type	Client port	Resolver	RCODE	Answers	Time delta
0xf0ba	example.com	A	51072	192.168.146.2:53	0	172.66.147.243; 104.20.23.154	0.286988101 s
0x4e43	example.com	AAAA	60932	192.168.146.2:53	0	2606:4700:10::6814:179a; 2606:4700:10::ac42:93f3	0.039519120 s

The query and response pairs were matched using the transaction ID and endpoint tuple. In both pairs, the response source was the configured resolver, 192.168.146.2. The measured time deltas were approximately 287 ms for the A transaction and 39.5 ms for the AAAA transaction.

7. Part E - DNS Generated by a Browser

The instructor-approved site used for the browser portion was example.com. The browser capture was performed with a port-53 filter so that DNS requests and responses could be inventoried without collecting unrelated application traffic.

```

Session  Actions  Edit  View  Help

(kali@kali)-[~/SBT-DF203-Lab7]
$ IFACE=eth0
sudo tshark -i "$IFACE" -f 'port 53' -a duration:40 -w /tmp/browser_dns.pcapng &
[2] 624751

(kali@kali)-[~/SBT-DF203-Lab7]
$ [sudo] password for kali:

[2] + suspended (tty input) sudo tshark -i "$IFACE" -f 'port 53' -a duration:40
-w /tmp/browser_dns.pcapn
(kali@kali)-[~/SBT-DF203-Lab7]
$ wait
ls -lh /tmp/browser_dns.pcapng
ls: cannot access '/tmp/browser_dns.pcapng': No such file or directory

(kali@kali)-[~/SBT-DF203-Lab7]
$

```

Figure 15: Initial browser DNS capture attempt showing the background sudo/TTY issue.

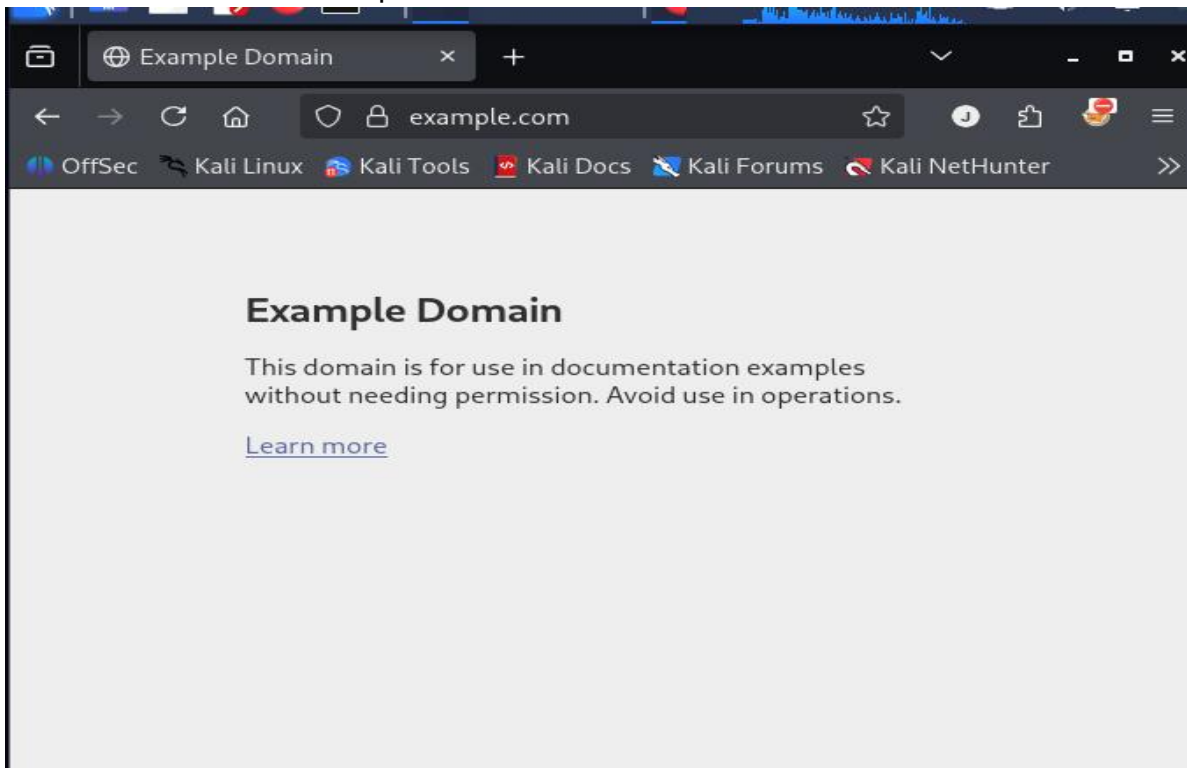


Figure 16: Browser opened to example.com for the controlled DNS-generation exercise.

```
(kali㉿kali)-[~/SBT-DF203-Lab7]
$ IFACE=eth0
sudo tshark -i "$IFACE" -f 'port 53' -a duration:40 -w /tmp/browser_dns.pcapng &
[3] 628157

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
6

[3] done sudo tshark -i "$IFACE" -f 'port 53' -a duration:40 -w /tmp/browser_dns.pcapn
(kali㉿kali)-[~/SBT-DF203-Lab7]
$ ls -lh /tmp/browser_dns.pcapng
-rw-r--r-- 1 root root 1.1K Sep 22 11:40 /tmp/browser_dns.pcapng

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ sudo mv /tmp/browser_dns.pcapng evidence/browser_dns.pcapng

(kali㉿kali)-[~/SBT-DF203-Lab7]
$ sudo chown kali:kali evidence/browser_dns.pcapng

(kali㉿kali)-[~/SBT-DF203-Lab7]
$
```

Figure 17: Corrected browser DNS capture completed successfully and evidence/browser_dns.pcapng was preserved.

```

Session Actions Edit View Help

(kali@kali)-[~/SBT-DF203-Lab7]
$ tshark -r evidence/browser_dns.pcapng -Y 'dns' -T fields \
-e frame.number \
-e frame.time_epoch \
-e ip.src \
-e udp.srcport \
-e ip.dst \
-e udp.dstport \
-e dns.id \
-e dns.flags.response \
-e dns.qry.name \
-e dns.qry.type \
-e dns.a \
-e dns.aaaa \
| tee reports/browser_dns_analysis.txt
1      1790091577.644211637      192.168.146.132 37121      192.168.146.2 53      0
x06ce  False  example.com 65
2      1790091577.644397317      192.168.146.132 40993      192.168.146.2 53      0
x09db  False  example.com 1
3      1790091577.644481254      192.168.146.132 40993      192.168.146.2 53      0
xd3dd  False  example.com 28
4      1790091577.655127726      192.168.146.2 53      192.168.146.132 37121 0
x06ce  True   example.com 65
5      1790091577.655128270      192.168.146.2 53      192.168.146.132 40993 0
x09db  True   example.com 1      172.66.147.243,104.20.23.154
6      1790091577.657582250      192.168.146.2 53      192.168.146.132 40993 0
xd3dd  True   example.com 28      2606:4700:10::ac42:93f3,2606:4700
:10::6814:179a

(kali@kali)-[~/SBT-DF203-Lab7]
$

```

Figure 18: Browser DNS field extraction showing A and AAAA queries and responses for example.com.

The browser-generated DNS evidence contained A and AAAA queries for example.com. The A response returned 172.66.147.243 and 104.20.23.154. The AAAA response returned 2606:4700:10::ac42:93f3 and 2606:4700:10::6814:179a. The presence of both IPv4 and IPv6 queries is a normal variation in modern DNS resolution.

An intermediate analysis attempt referenced the field dns.rdata, which TShark reported as invalid on this installation. The analysis was then corrected to use supported DNS fields such as dns.a and dns.aaaa. The corrected extraction produced the browser DNS results shown above.

8. Part F - Correlation of DNS Answers with Subsequent Connections

The required DNS A-answer extraction produced one record for example.com containing two IPv4 addresses: 172.66.147.243 and 104.20.23.154. The subsequent TCP-destination extraction produced no rows.

This absence does not demonstrate that the browser made no connection. The browser PCAPNG was created with the capture filter '-f port 53', which restricts the recorded traffic to DNS packets. Consequently, later TCP/HTTPS SYN packets were outside the capture scope and could not be correlated from that file. The lab instructions explicitly require the analyst to explain mismatches using factors such as capture scope/duration, multiple answers, IPv6, caching, proxies and CDNs. In this case, the principal evidence-based limitation is the DNS-only capture filter.

Part F item	Observed result	Forensic interpretation
DNS A answer	172.66.147.243; 104.20.23.154	Two IPv4 answers observed for example.com.
TCP destination extraction	No rows	No TCP packets were captured because the PCAPNG was filtered to port 53.
Connection correlation	Not directly observable	Requires a capture that includes the subsequent TCP/HTTPS traffic.

9. Section 15 - Required Findings Worksheet / Analysis

The following findings answer the Section 15 worksheet using the captured evidence and command outputs produced during the practical.

Required finding	Finding / analysis
Configured resolver IP	192.168.146.2. This resolver is shown in the dig SERVER lines and in the TShark DNS query/response endpoints.
Client source port	Fresh capture: 51072 for the A query and 60932 for the AAAA query. Browser capture: 37121 and 40993 were observed.
Resolver destination port	UDP port 53.
Transaction ID	Fresh A transaction: 0xf0ba. Fresh AAAA transaction: 0x4e43. The transaction IDs match their corresponding responses.
Query name and type	example.com, type A in the first transaction and type AAAA in the second.
Response code	0 / NOERROR for the observed successful A and AAAA responses.
Answer IP(s)	A: 172.66.147.243 and 104.20.23.154. AAAA: 2606:4700:10::ac42:93f3 and 2606:4700:10::6814:179a.
TTL	5 seconds for each A and AAAA answer observed in the fresh correlation. The TTL represents a cache lifetime.
Query-response time delta	A: approximately 0.286988101 seconds. AAAA: approximately 0.039519120 seconds. Browser A and AAAA responses were also returned within milliseconds of their queries.
Subsequent connection correlation	Not directly demonstrated. The browser capture contained DNS-only traffic because it used a port-53 capture filter, so no TCP SYN packets were available for the required later-connection comparison.
Normal variations observed	Multiple A answers, multiple AAAA answers, IPv4 and IPv6 resolution, short TTL values, and initial timeout messages on the MX and NS commands followed by successful responses. These observations are consistent with normal DNS

variation and capture/resolver timing rather than, by themselves, evidence of an attack.

10. Limitations and Operational Notes

- The MX and NS dig commands displayed an initial communication timeout before producing successful NOERROR responses. The final successful responses were used for the findings.
- The first fresh-capture workflow encountered a permission-denied error when root-owned TShark attempted to write directly into the evidence directory. The corrected /tmp → evidence workflow preserved the resulting capture.
- The browser DNS capture was intentionally limited to port 53, so it cannot establish a later TCP/HTTPS connection to the returned IP addresses.
- An initial TShark field extraction attempted to use dns.rdata, which was unsupported in the installed TShark version. Supported fields were used for the final analysis.
- Part G (DNS analysis within SMTP evidence) was not performed because the lab instructions mark that section as optional.

11. Evidence Inventory and Reproducibility

Artifact	Role / contents
evidence/dig_dns.pcap	Supplied DNS capture; preserved original
working/dig_dns_working.pcap	Working analysis copy of supplied capture
evidence/fresh_dig_dns.pcapng	Fresh controlled DNS capture
evidence/browser_dns.pcapng	Browser-generated DNS capture
reports/dns_capture_hashes.txt	Hash record for supplied capture and working copy
reports/fresh_dns_sha256.txt	Hash record for fresh DNS capture
reports/dig_example_A.txt	A-record dig output
reports/dig_example_AAAA.txt	AAAA-record dig output
reports/dig_example_MX.txt	MX-record dig output
reports/dig_example_NS.txt	NS-record dig output
reports/dig_example_short.txt	Short A-record output
reports/dns_queries.tsv	Extracted DNS query fields
reports/dns_responses.tsv	Extracted DNS response fields
reports/dns_correlation.tsv	Matched query/response transactions
reports/browser_dns_analysis.txt	Browser DNS field analysis
reports/dns_A_answers.tsv	Browser DNS A-answer extraction
reports/subsequent_tcp_destinations.tsv	TCP destination extraction; no rows because capture was DNS-only

12. Conclusion

The examination established a reproducible DNS-forensics workflow for normal-traffic baseline analysis. The configured resolver was identified as 192.168.146.2, the supplied capture and working copy were verified by matching SHA-256 hashes, and controlled dig queries demonstrated successful A, AAAA, MX and NS resolution for example.com.

Packet-level analysis confirmed that DNS transactions can be matched through transaction IDs and endpoint tuples. The fresh A and AAAA responses returned two addresses each with a TTL of 5 seconds, while the browser capture demonstrated that a normal web visit can generate both IPv4 and IPv6 DNS lookups.

The Part F correlation limitation is explicitly documented: because the browser capture was restricted to DNS port 53, the later TCP connection was not captured and therefore cannot be claimed from that evidence. This distinction preserves the forensic boundary between what the evidence demonstrates and what it does not. The optional SMTP section was intentionally omitted.

Overall, the completed evidence supports the lab objective of establishing a normal DNS traffic baseline suitable for later forensic comparison, including the spoofing-analysis work introduced in the subsequent lab.